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Self-Watering Planter with Soil Moisture Feedback Control

ABSTRACT

A planter assembly includes a soil-filled upper basin, a water reservoir positioned beneath the basin, a wicking column connecting the reservoir to the soil, and a moisture sensor coupled to a small pump. The pump draws water from an external supply into the reservoir only when the sensor detects that soil moisture has fallen below a configured threshold, reducing both overwatering and manual maintenance.

BACKGROUND

Conventional self-watering planters rely on constant capillary wicking without feedback, which can oversaturate soil for moisture-sensitive plants or fail to deliver adequate water during periods of high evaporation. A feedback-controlled watering mechanism can adapt to actual soil conditions rather than a fixed wicking rate.

SUMMARY

In one embodiment, a capacitive moisture sensor is embedded in the soil basin at a fixed depth and coupled to a microcontroller. When the measured moisture falls below a configured threshold, the microcontroller activates a small pump that transfers water from an external supply line into the reservoir, which then wicks upward into the soil through a fabric column. The microcontroller logs each activation event to support later adjustment of the threshold.

CLAIMS

1. A planter apparatus comprising: a soil basin; a water reservoir positioned beneath the soil basin; a wicking column fluidly connecting the reservoir to the soil basin; a moisture sensor positioned within the soil basin; and a pump configured to add water to the reservoir in response to a signal from the moisture sensor. 2. The apparatus of claim 1, wherein the moisture sensor is a capacitive sensor. 3. The apparatus of claim 1, further comprising a microcontroller configured to log each activation of the pump.